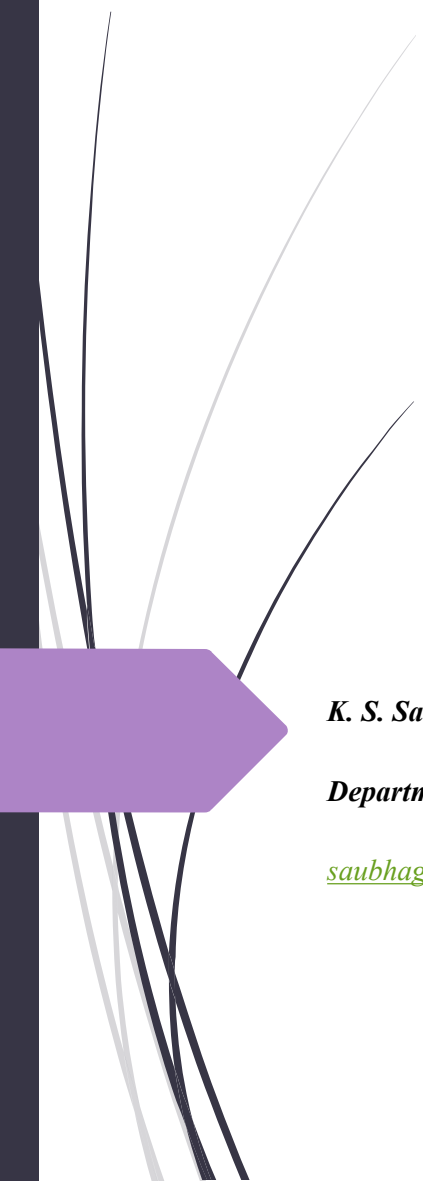




ST 3007

Operational Research

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Integer Programming – Branching and Bounding Method

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Chapter 2

Chapter Summary

- **Branching and Bounding Method**
- **B&B Algorithm to Maximization problem**

Branching and Bounding Method

- Mixed Integer Programming (MIP) problems cannot be solved using the cutting plane method, since it assumes that all the variables should be integer values.
- To solve MIP problems, we can use the Branching and Bounding (**B&B**) method.
- The B&B method is not a solution technique specifically limited to Integer Programming (IP) problems. It is a solution approach that can be applied to a number of different types of problems.
- The B&B approach is based on the principle that the total set of feasible solutions can be partitioned into smaller subsets of solutions. These smaller subsets can then be evaluated systematically until the best solution is found.
- When the B&B approach is applied to an IP problem, it is used in conjunction with the normal non-integer solution approach.

Branching – B&B Method

- ▶ If the solution to the Linear Programming (LP) problem contains decision variables with non-integer values, then the solution space is reduced by introducing constraints with respect to one of those decision variables (branching variable).
- ▶ When selecting the variable for branching, choose the variable which has the largest fractional part. If there is a **tie**, select the variable that has **the higher objective function coefficient**.
- ▶ If the value of the branching decision variable X_i is b (non-integer) then select “ k ” the largest integer less than b .
- ▶ Thus, a pair of branches are formulated such that X_i is upper bounded by k and lower bounded by $k + 1$:

$$X_i \leq k \text{ and } X_i \geq k + 1$$

- ▶ Now, two sub-problems can be created.

Sub problem j : most recent problem plus the new constraint $X_i \leq k$

Sub problem j+1 : most recent problem plus the new constraint $X_i \geq k + 1$

Bounding – B&B Method

- For each new sub-problem, obtain their optimal solutions by applying the simplex method (or the dual simplex method when reoptimizing) to its LP relaxation.
- Upper bound (Z_u) is the value obtained from the resulting optimal solution.
- Lower bound (Z_l) is the value obtained from the rounded down (for maximization) or rounded up integer solutions (for minimization).
- Current best lower Bound is the best lower bound (highest in maximization and lowest in minimization problems) among the lower bounds of all the fathomed nodes (will discuss in next slide). Initially, it is assumed for the root node as below.

Maximization problem : $-\infty$

Minimization problem : $+\infty$

Fathoming - B&B Method

- ▶ A problem (node) is said to be fathomed if any one of the following three conditions is true:
 - 1. The values of the decision variables of the problem are integer.**
 - 2. The upper bound of the problem which has non-integer values for its decision variable is not greater than the current best lower bound.**
 - 3. The problem has an infeasible solution.**
- ▶ If a node is fathomed, further branching from the node is unnecessary.

B&B Algorithm to Maximization problem

- ▶ The steps of the branch and bound method for determining an optimal integer solution for a maximization model can be summarized as follows.
 - **Step 1:** Find the optimal solution to the LP problem with the integer restrictions relaxed (Initially set the current best lower bound as $-\infty$).
 - **Step 2:** Check whether the problem has an integer solution. If yes, the current solution is the optimal solution. Otherwise, let the relaxed solution be the upper bound and the rounded-down integer solution be the lower bound (current best lower bound).
 - **Step 3:** Select the variable with the greatest fractional part for branching (If there is a tie, select the variable that has the higher objective function coefficient). Create two new constraints for this variable reflecting the partitioned integer values.

$$\begin{aligned} X_k &\leq \text{Integer part of } X_k \\ X_k &\geq \text{Integer part of } X_k + 1 \end{aligned}$$

B&B Algorithm to Maximization problem

- **Step 4:** Create two new nodes, one for the $\leq$ constraint and one for the $\geq$ constraint. Solve the relaxed linear programming models with the new constraint added at each of these nodes.
- **Step 5:** If any one of the new subproblems has an infeasible solution, fully integer values for the decision variables or upper bound is not greater than the current best lower bound, the corresponding node is fathomed. Further, if a new node has integer values for decision variables, update the current best lower bound as the lower bound of that node if it's lower bound is greater than the previous current best lower bound.
- **Step 6:** Are all the terminal nodes fathomed? If yes, go to step 7, otherwise, go to step 3 (only consider unfathomed nodes).
- **Step 7:** Select the solution of the problem with respect to the fathomed node whose lower bound is equal to the current best lower bound as the optimal solution.

Example: B&B method

- Consider the following Integer Programming problem.

$$\text{Maximize } Z = 10x_1 + 20x_2$$

Subject to:

$$6x_1 + 8x_2 \leq 48$$

$$x_1 + 3x_2 \leq 12$$

$$x_1, x_2 \geq 0 \text{ and integers}$$

- B&B method can be used to solve both MIP as well as PIP problems.
- Let's use the B&B algorithm to find the optimal solution to the above IP problem.

Example: B&B method Solution

- We begin the branch and bound method by first solving the problem as a regular linear programming model without integer restrictions (i.e., the integer restrictions are relaxed).

$$P_0 \Rightarrow \begin{cases} \text{Max } Z = 10x_1 + 20x_2 \\ 6x_1 + 8x_2 \leq 48 \\ x_1 + 3x_2 \leq 12 \\ x_1, x_2 \geq 0 \end{cases}$$

Solⁿ

$$x_1 = 24/5 = 4 \frac{4}{5}$$

$$x_2 = 12/5 = 2 \frac{2}{5}$$

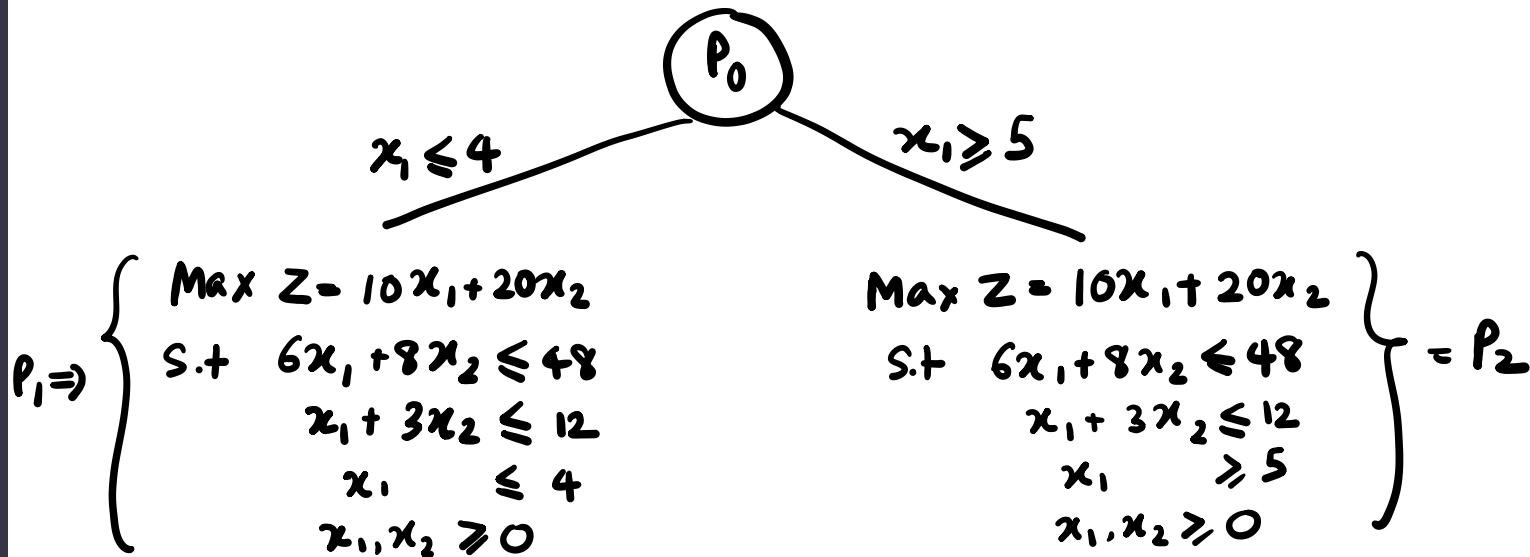
$$Z_u = 96 \quad Z_L = 10(4) + 20(2) = 80$$

Current best lower bound } = 80

- Since x_1 has the highest fraction, we select x_1 as the branching variable. Two branches are $x_1 \leq 4$ and $x_1 \geq 5$.
- The branch and bound method employ a diagram consisting of nodes and branches as a framework for the solution process.

Example: B&B method Solution

- Subproblems derived from branching are given below.



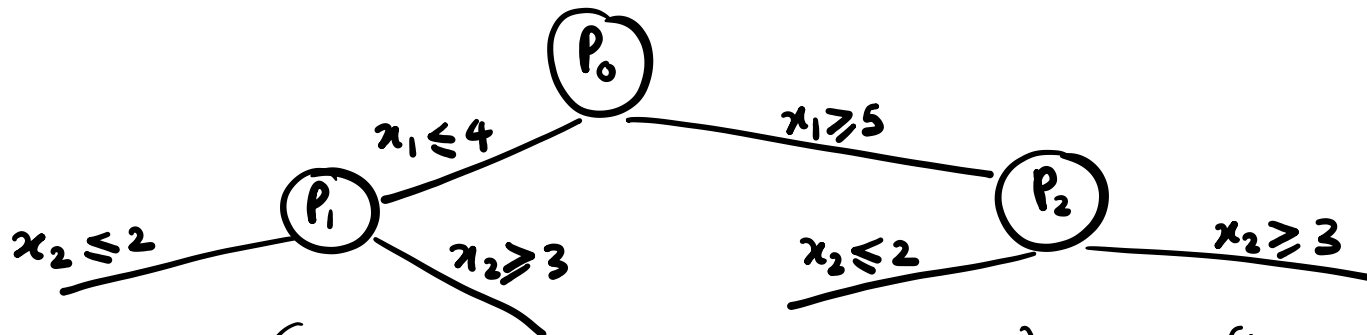
- Subproblems are solved using either the simplex method or the dual simplex method.

$$\begin{array}{l}
 P_1 \text{ solution} \Rightarrow \\
 x_1 = 4 \\
 x_2 = 8/3 = 2\frac{2}{3} \\
 Z_u = 93\frac{1}{3} \quad Z_L = 80
 \end{array}$$

$$\begin{array}{l}
 P_2 \text{ solution} \Rightarrow \\
 x_1 = 5 \\
 x_2 = 9/4 = 2\frac{1}{4} \\
 Z_u = 95 \quad Z_L = 90
 \end{array}$$

Example: B&B method Solution

➤ We will continue with creating subproblems from each node.



$$P_3 \left\{ \begin{array}{l} \text{Max } Z = 10x_1 + 20x_2 \\ 6x_1 + 8x_2 \leq 48 \\ x_1 + 3x_2 \leq 12 \\ x_1 \leq 4 \\ x_2 \leq 2 \\ x_1, x_2 \geq 0 \end{array} \right.$$

$$x_1 = 4 \quad x_2 = 2 \\ Z_u = Z_l = 80 \text{ [Fathomed]}$$

$$P_4 \left\{ \begin{array}{l} \text{Max } Z = 10x_1 + 20x_2 \\ 6x_1 + 8x_2 \leq 48 \\ x_1 + 3x_2 \leq 12 \\ x_1 \leq 4 \\ x_2 \geq 3 \\ x_1, x_2 \geq 0 \end{array} \right.$$

$$x_1 = 3 \quad x_2 = 3 \\ Z_u = Z_l = 90 \text{ [Fathomed]}$$

This becomes
current best
lower bound

$$P_5 \left\{ \begin{array}{l} \text{Max } Z = 10x_1 + 20x_2 \\ 6x_1 + 8x_2 \leq 48 \\ x_1 + 3x_2 \leq 12 \\ x_1 \geq 5 \\ x_2 \leq 2 \\ x_1, x_2 \geq 0 \end{array} \right.$$

$$x_1 = 5\frac{1}{3} \quad x_2 = 2 \\ Z_u = 93\frac{1}{3} \quad Z_l = 90 \text{ [not fathomed]}$$

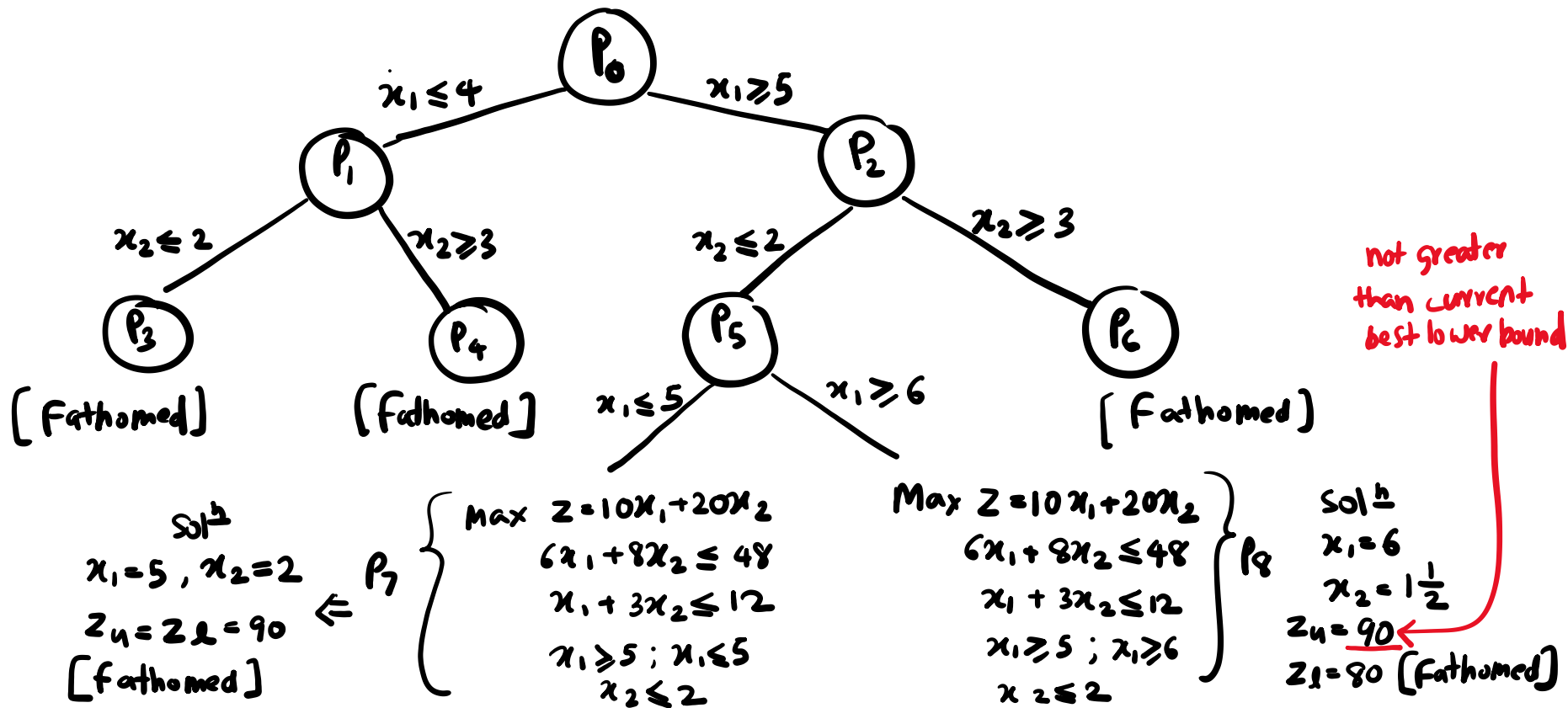
$$P_6 \left\{ \begin{array}{l} \text{Max } Z = 10x_1 + 20x_2 \\ 6x_1 + 8x_2 \leq 48 \\ x_1 + 3x_2 \leq 12 \\ x_1 \geq 5 \\ x_2 \geq 3 \\ x_1, x_2 \geq 0 \end{array} \right.$$

Infeasible Solution
[Fathomed]

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Example: B&B method Solution

- No new node from all the fathomed nodes. Only one node to continue.



Example: B&B method Solution

- ▶ Once all the nodes are fathomed, we don't have any new nodes to branch out.
- ▶ Then we choose the optimal solution from the fathom nodes whose lower bound is equal to the current best lower bound.
- ▶ Here we have two fathomed nodes with a lower bound equaling to the current best lower bound.
- ▶ Hence, we have alternate optimal solutions.

$x_1 = 3; x_2 = 3; z = 90$ and $x_1 = 5; x_2 = 2$ and $z = 90$

B&B Method – Remarks

- Mixed integer linear programming problems can also be solved using the branch and bound method. The same basic steps that were applied to the total integer model in the previous section are used for a mixed integer model with only a few differences.
 - First, at the root node only those variables with integer restrictions are rounded down to achieve the lower bound.
 - Second, in determining which variable to branch from, we select the greatest fractional part from among only those variables that must be an integer.
- For a minimization model, relaxed solutions are rounded up, and upper and lower bounds are reversed.

Thank You!

Next Chapter - Binary Integer Programming